NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Unusual Does Not Mean Impossible

AP Statistics · Topic 4.11 (CED 2.10) · B: 2026-11-06 · E: 2026-11-30

[STAR] TODAY'S PROBLEM

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: "Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies."

Analyst: "A result can be unusual without being impossible; compare 58 with the model's mean and standard deviation."

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE — before the video (5 min)

Whose statement is more defensible, the coach's or the analyst's? Name one quantity you would check.

[BOOK] CENTER, SPREAD, AND UNUSUAL COUNTS (keep on the page)

For $X \sim \text{Bin}(n, p)$, $\mu_X = np$ and $\sigma_X = \sqrt{np(1-p)}$; both use count units. Standardize with $z = \dfrac{x - \mu_X}{\sigma_X}$. A normal approximation is reasonable when both $np \geq 10$ and $n(1-p) \geq 10$. A count far from the mean may be unusual, but every value from 0 through n still has positive binomial probability when $0 < p < 1$.

Finish after the video:

DOK 1 **(a)** State n and p for X and write the formulas for the binomial mean and standard deviation.

DOK 2 **(b)** Without calculating yet, interpret the model's mean and standard deviation in the context of many sets of 100 free throws.

DOK 3 **(c)** Compute μ_X, σ_X, and the z-score of 58. Check np and $n(1-p)$ to decide whether a normal approximation is reasonable. Is 58 unusual? Adjudicate the two statements, distinguish "unusual" from "impossible," and explain what the coach's claim could mislead a reader into believing.

Frame: The value 58 is ____________ standard deviations ____________ the mean, so it is ____________.

Frame: The coach confuses ____________________ with __________________, which could make a reader believe ____________________________________.

EXIT — turn this in

One thing the video changed about my first take: ____